Customer Behavior Analysis Report

1. Objective

The goal of this project is to analyze customer transaction data to identify purchasing patterns, segment customers based on their behavior, and detect potential churn risks. The insights will help improve customer engagement and business strategy.

2. Dataset Overview

The dataset contains **250,000 customer transaction records** with the following key features:

- Customer ID
- Purchase Date
- Product Category
- Product Price & Quantity
- Total Purchase Amount
- Payment Method
- Customer Age, Gender
- Churn

This dataset provides a comprehensive view of customer purchasing behavior.

3. Data Cleaning & Preparation

The following steps were performed:

- Removed duplicates and checked for missing values
- Converted *Purchase Date* to datetime format
- Created additional features like Month and Year
- Verified consistency in numerical and categorical data

4. Customer Segmentation (RFM Analysis)

Customers were segmented using:

- **Recency (R)** → Days since last purchase
- **Frequency (F)** → Number of purchases
- **Monetary (M)** → Total spending

Segments Created:

- **High Value Customers** → High spending customers
- **Loyal Customers** → Frequent buyers

- **Low Value Customers** → Low spending and less frequent

5. Key Insights

Customer Distribution

- Majority of customers fall under the **Low Value segment**
- A smaller group contributes to **high revenue (High Value customers)**

Spending Behavior

- High Value customers contribute significantly to total revenue
- Loyal customers purchase frequently but may not always spend the most

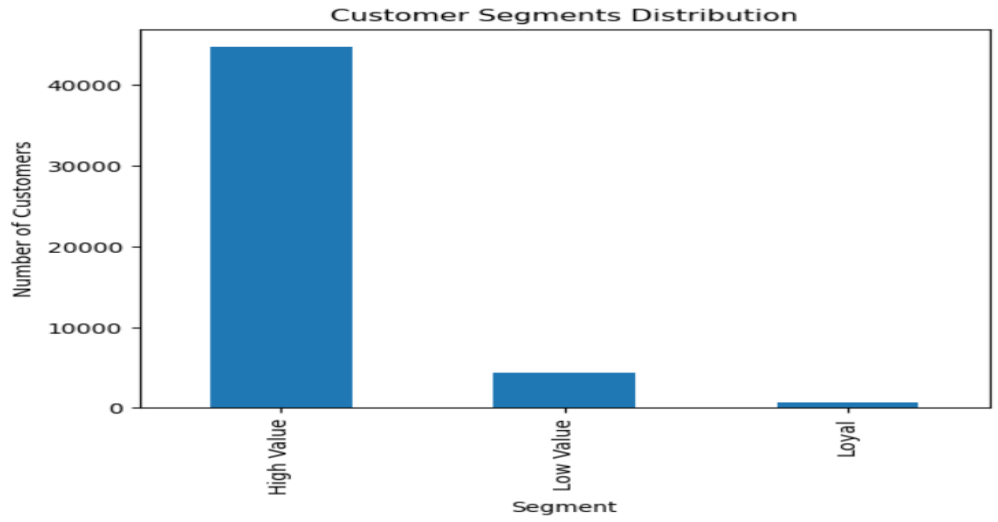
Product Trends

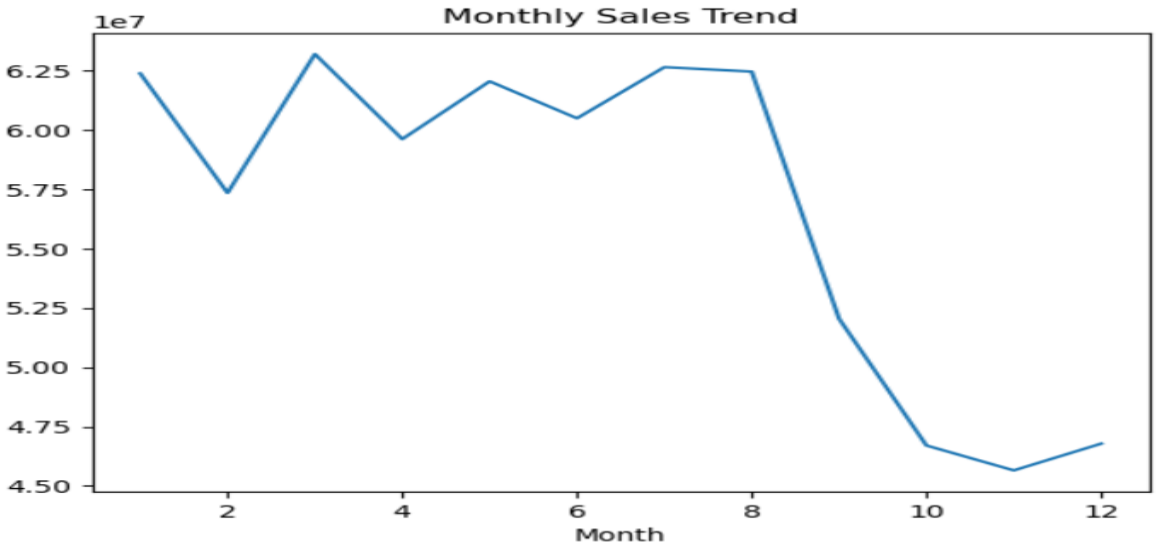
- Certain product categories dominate sales (e.g., Home, Electronics, etc.)
- Customers show repeat behavior in preferred categories

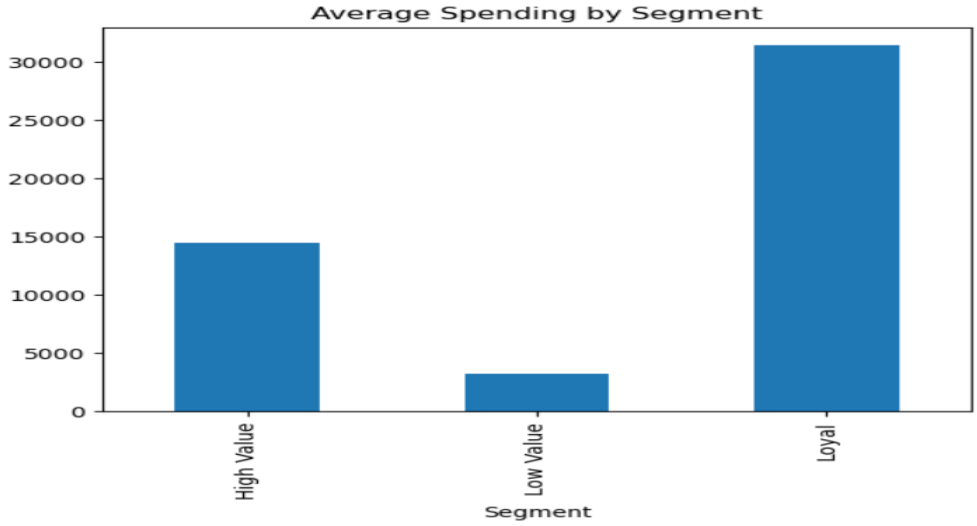
Churn Insights

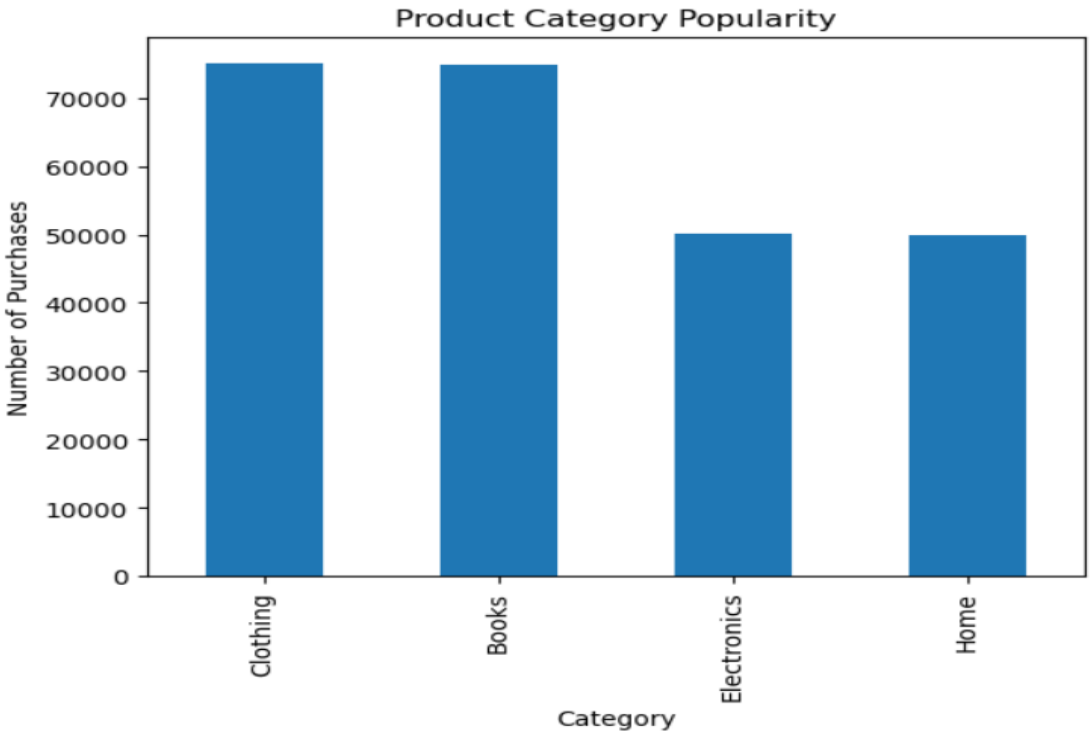
- Some customers show low activity (high recency), indicating churn risk
- Retention strategies are needed for such customers

6. Visualizations




Monthly Sales Trend
1e7
6.25
6.00
5.75
5.50
5.25
5.00
4.75
4.50
2
4
6
8
10
12
Month


Average Spending by Segment
30000
25000
20000
15000
10000
5000
0
High Value
Low Value
Loyal
Segment



- Customer segment distribution
- Monthly sales trends
- Product category popularity
- Spending by segment

7. Recommendations

1. **Reward High-Value Customers**
   Provide loyalty programs, exclusive discounts, and early access to products.
2. **Increase Engagement of Low-Value Customers**
   Offer targeted promotions and discounts to encourage more purchases.
3. **Retain At-Risk Customers**
   Identify inactive customers and send personalized offers or reminders.
4. **Focus on Popular Categories**
   Invest more in high-performing product categories for better returns.
5. **Use Personalized Marketing**
   Recommend products based on customer purchase history to improve conversions.

8. Conclusion

This analysis highlights that a small group of customers contributes significantly to revenue, while a large portion remains low-value. By focusing on customer segmentation and targeted

marketing strategies, businesses can improve retention, increase revenue, and enhance customer satisfaction.